A7q3 comsol

Report date

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1 Global Definitions

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GLOBAL SETTINGS

Name	A7q3 comsol.mph
Path	E:\Comsol codes\a7q3_comsol.mph
Version	COMSOL Multiphysics 6.0 (Build: 318)

USED PRODUCTS

COMSOL Multiphysics

COMPUTER INFORMATION

CPU	Intel64 Family 6 Model 142 Stepping 12, 2 cores
Operating system	Windows 10

1.1 PARAMETERS

PARAMETERS 1

Name	Expression	Value	Description
cmax	3[mol/m^3]	3 mol/m³	
D	1e-5[m^2/s]	1E−5 m²/s	
k_m	5[m/s]	5 m/s	
c1	0[mol/m^3]	0 mol/m³	
R	0[mol/(m^3*s)]	0 mol/(m³·s)	
a	1000	1000	

2 Component 1

2.1 DEFINITIONS

2.1.1 Variables

Variables 1

SELECTION

Geometric entity level	Entire model

Name	Expression	Unit	Description
c0	cmax*exp(-a*(x/0.4[mm])^2)	mol/m³	

2.1.2 Coordinate Systems

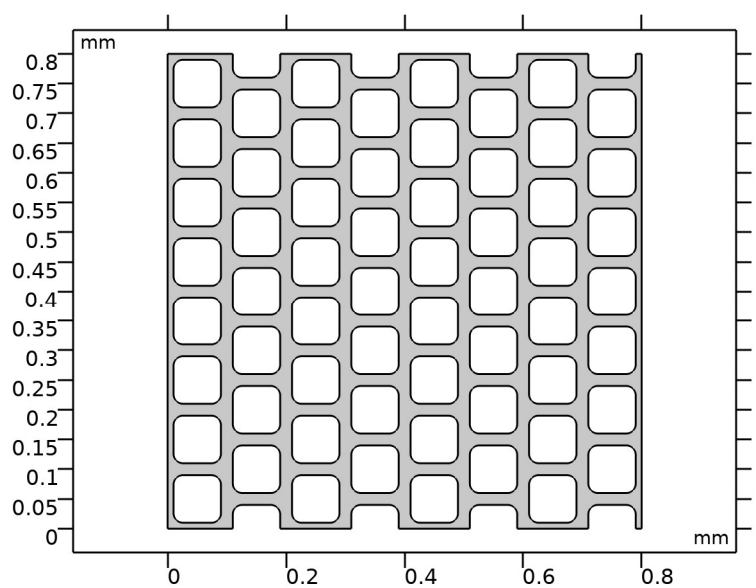
Boundary System 1

Coordinate system type	Boundary system
Tag	sys1

COORDINATE NAMES

First	Second	Third
t1	n	to

2.2 GEOMETRY 1

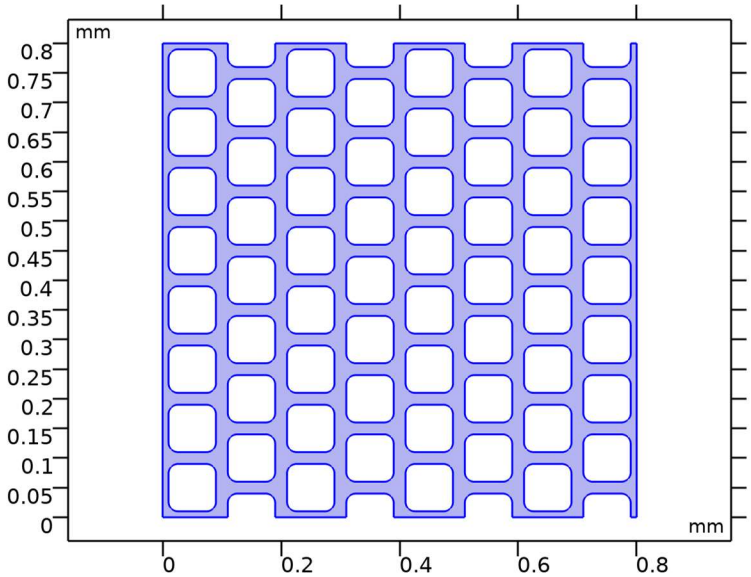


Geometry 1

UNITS

Length unit	mm
Angular unit	deg

2.3 TRANSPORT OF DILUTED SPECIES



Transport of Diluted Species

EQUATIONS

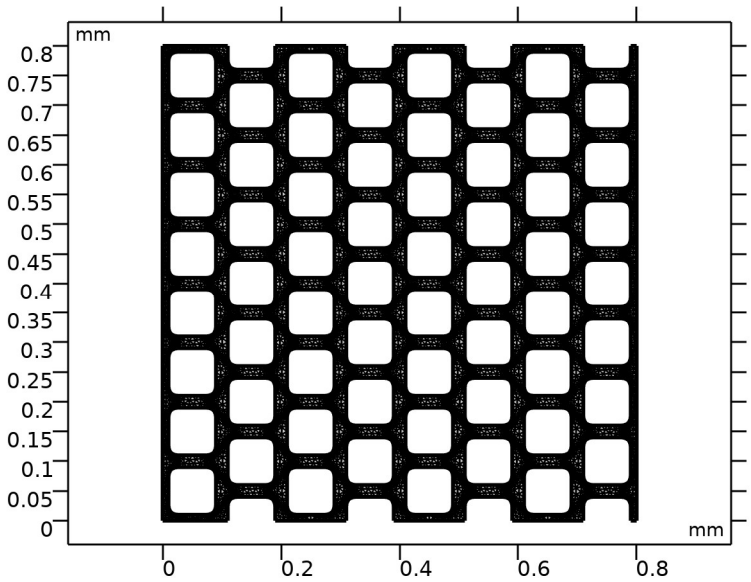
$$\frac{\partial c_i}{\partial t} + \nabla \cdot \mathbf{J}_i + \mathbf{u} \cdot \nabla c_i = R_i$$

$$\mathbf{J}_i = -D_i \nabla c_i$$

FEATURES

Name	Level
Transport Properties 1	Domain
No Flux 1	Boundary
Initial Values 1	Domain
Concentration 1	Boundary
Reactions 1	Domain
Flux 1	Boundary

2.4 MESH 1



Mesh 1

3 Study 1

COMPUTATION INFORMATION

Computation time	22 s

3.1 TIME DEPENDENT

Times	Unit
range(0,1,20)	s

STUDY SETTINGS

Description	Value
Include geometric nonlinearity	Off

STUDY SETTINGS

Description	Value
Output times	{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20}

PHYSICS AND VARIABLES SELECTION

Physics interface	Discretization
Transport of Diluted Species (tds)	physics

MESH SELECTION

Geometry	Mesh
Geometry 1 (geom1)	mesh1

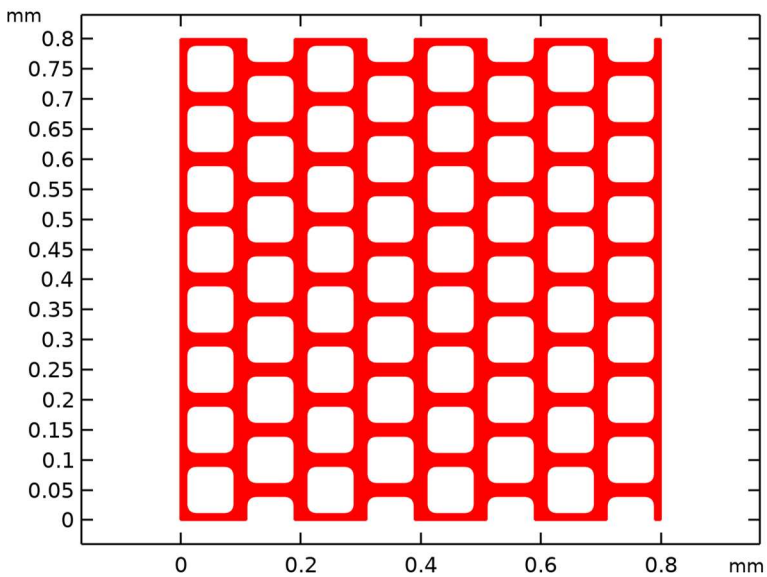
4 Results

4.1 DATASETS

4.1.1 Study 1/Solution 1

SOLUTION

Description	Value
Solution	Solution 1
Component	Component 1 (comp1)



Dataset: Study 1/Solution 1

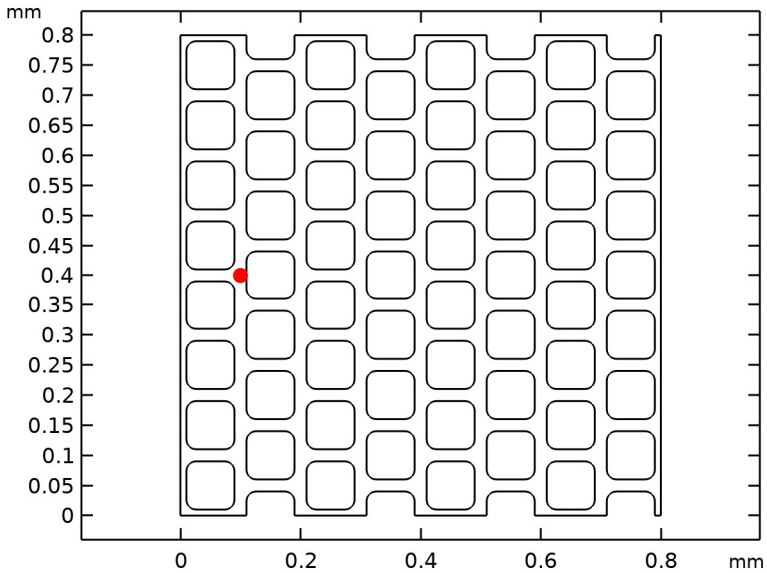
4.1.2 Cut Point 2D 1

DATA

Description	Value
Dataset	Study 1/Solution 1

POINT DATA

Description	Value
Entry method	Coordinates
X	0.1
Y	0.4



Dataset: Cut Point 2D 1

4.1.3 Cut Point 2D 2

DATA

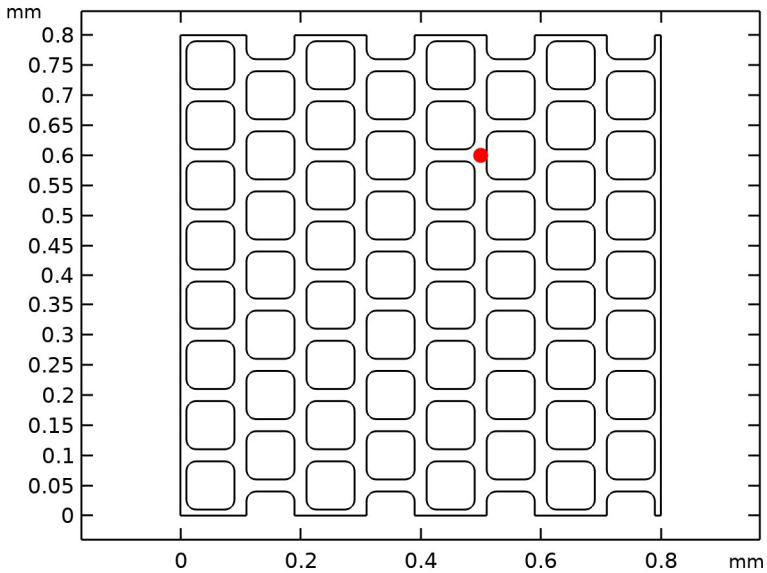
Description	Value
Dataset	Study 1/Solution 1

POINT DATA

Description	Value
Entry method	Coordinates
X	0.5
Y	0.6

ADVANCED

Description	Value
Point number variable	cpt1n



Dataset: Cut Point 2D 2

4.1.4 Cut Point 2D 3

DATA

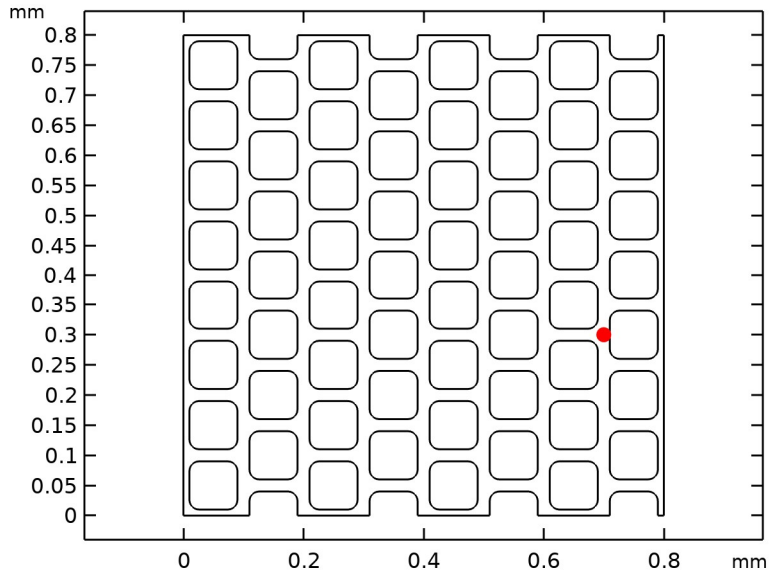
Description	Value
Dataset	Study 1/Solution 1

POINT DATA

Description	Value
Entry method	Coordinates
X	0.7
Y	0.3

ADVANCED

Description	Value
Point number variable	cpt1n



Dataset: Cut Point 2D 3

4.1.5 Cut Line 2D 1

DATA

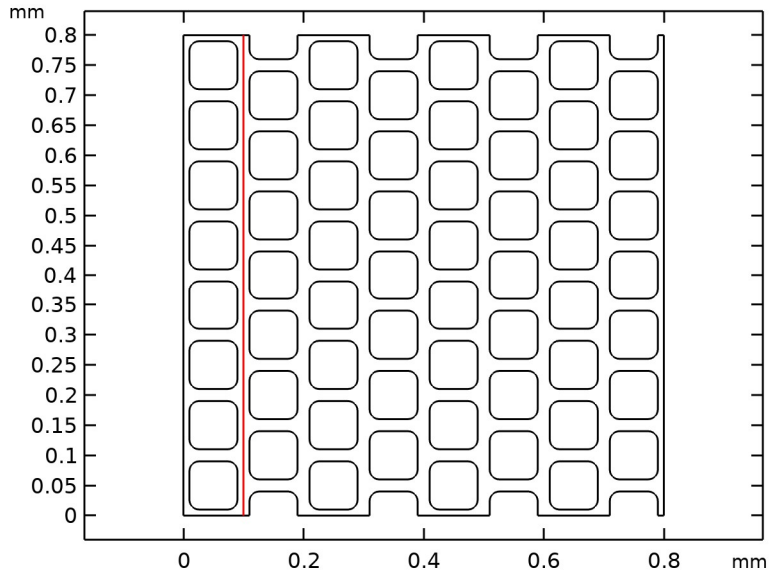
Description	Value
Dataset	Study 1/Solution 1

LINE DATA

Description	Value
Line entry method	Two points
Points	{{0.1, 0}, {0.1, 0.8}}

ADVANCED

Description	Value
Space variable	cln1x
Normal variables	{cln1nx, cln1ny}
Tangent variables	{cln1tx, cln1ty}



Dataset: Cut Line 2D 1

4.1.6 Cut Line 2D 2

DATA

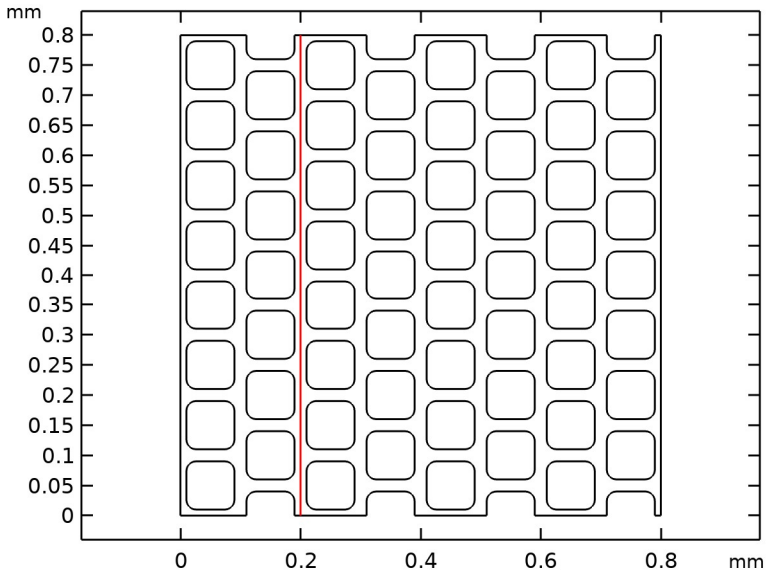
Description	Value
Dataset	Study 1/Solution 1

LINE DATA

Description	Value
Line entry method	Two points
Points	{{0.2, 0}, {0.2, 0.8}}

ADVANCED

Description	Value
Space variable	cln1x
Normal variables	{cln1nx, cln1ny}
Tangent variables	{cln1tx, cln1ty}



Dataset: Cut Line 2D 2

4.1.7 Cut Line 2D 3

DATA

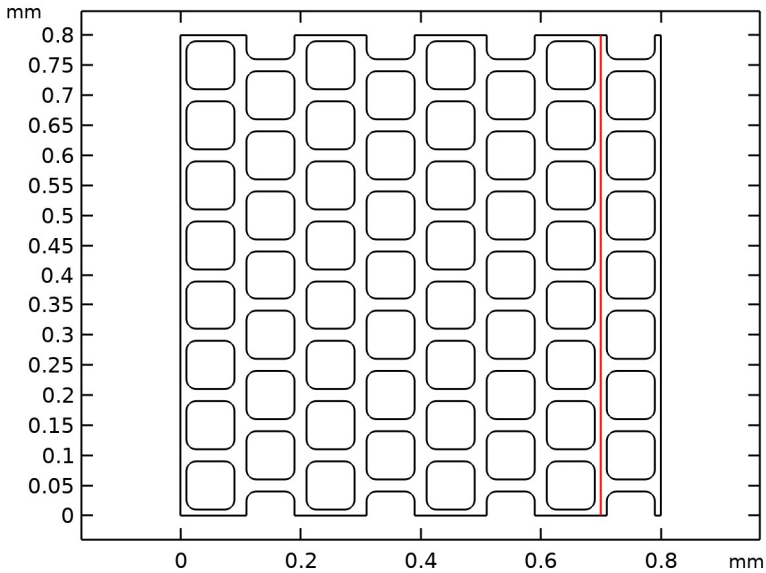
Description	Value
Dataset	Study 1/Solution 1

LINE DATA

Description	Value
Line entry method	Two points
Points	{{0.7, 0}, {0.7, 0.8}}

ADVANCED

Description	Value
Space variable	cln1x
Normal variables	{cln1nx, cln1ny}
Tangent variables	{cln1tx, cln1ty}



Dataset: Cut Line 2D 3

4.1.8 Cut Line 2D 4

DATA

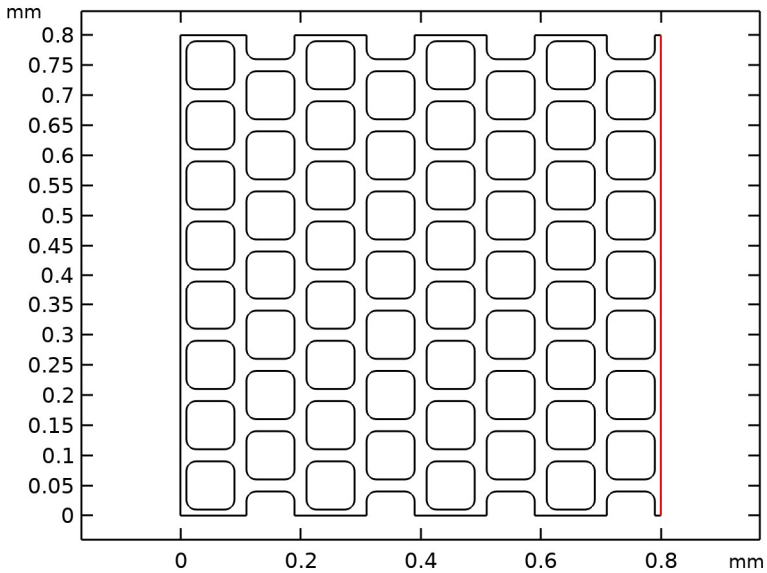
Description	Value
Dataset	Study 1/Solution 1

LINE DATA

Description	Value
Line entry method	Two points
Points	{{0.8, 0}, {0.8, 0.8}}

ADVANCED

Description	Value
Space variable	cln1x
Normal variables	{cln1nx, cln1ny}
Tangent variables	{cln1tx, cln1ty}



Dataset: Cut Line 2D 4

4.2 DERIVED VALUES

4.2.1 Point Evaluation 1

OUTPUT

Evaluated in	Table 1

DATA

Description	Value
Dataset	Cut Point 2D 1

EXPRESSIONS

Expression	Unit	Description
c	mol/m^3	Concentration

4.2.2 Point Evaluation 2

OUTPUT

Evaluated in	Table 1

DATA

Description	Value
Dataset	Cut Point 2D 2

EXPRESSIONS

Expression	Unit	Description

Expression	Unit	Description
c	mol/m^3	Concentration

4.2.3 Point Evaluation 3

OUTPUT

Evaluated in	Table 1

DATA

Description	Value
Dataset	Cut Point 2D 3

EXPRESSIONS

Expression	Unit	Description
c	mol/m^3	Concentration

4.3 TABLES

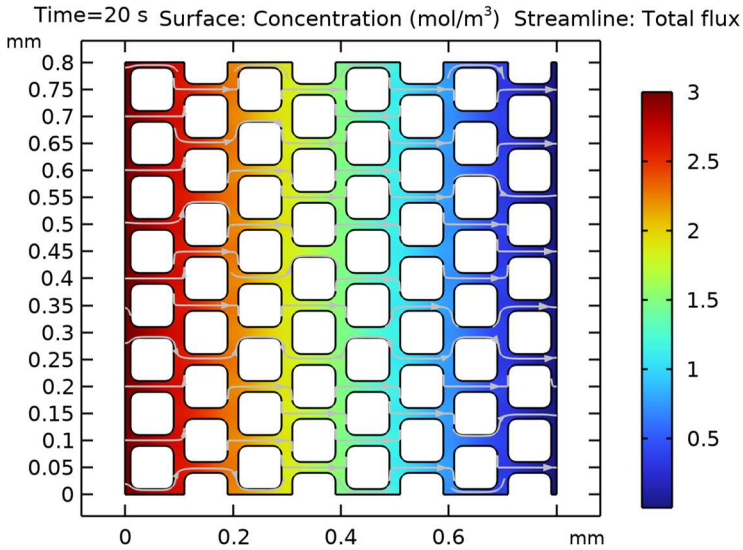
4.3.1 Table 1

Point Evaluation 1

Time (s)	Concentration (mol/m^3), Point: (0.1, 0.4)	Concentration (mol/m^3), Point: (0.5, 0.6)	Concentration (mol/m^3), Point: (0.7, 0.3)	Concentration (mol/m^3), Point: (0.1, 0.4)	Concentration (mol/m^3), Point: (0.1, 0.4)	Concentration (mol/m^3), Point: (0.1, 0.4)	Concentration (mol/m^3), Point: (0.1, 0.4)	Concentration (mol/m^3), Point: (0.5, 0.6)	Concentration (mol/m^3), Point: (0.1, 0.4)
20.000	2.6660	1.1494	0.39094	2.6660	2.6660	2.6660	2.6660	1.1494	2.6660

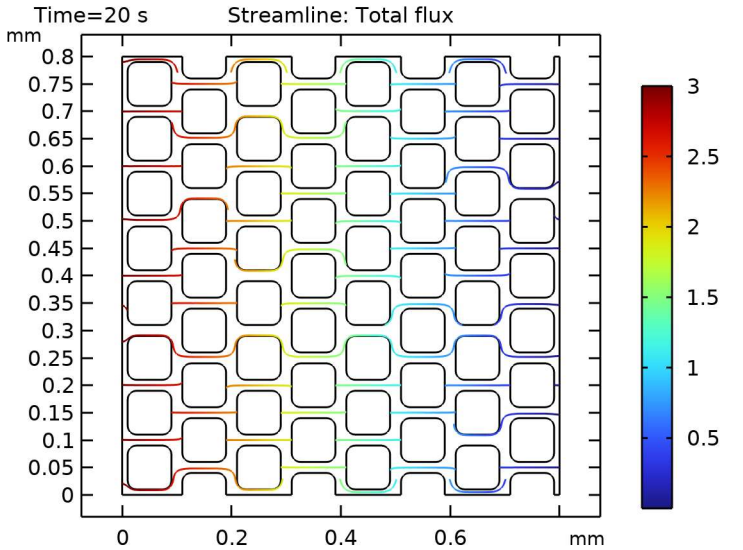
4.4 PLOT GROUPS

4.4.1 Concentration (tds)



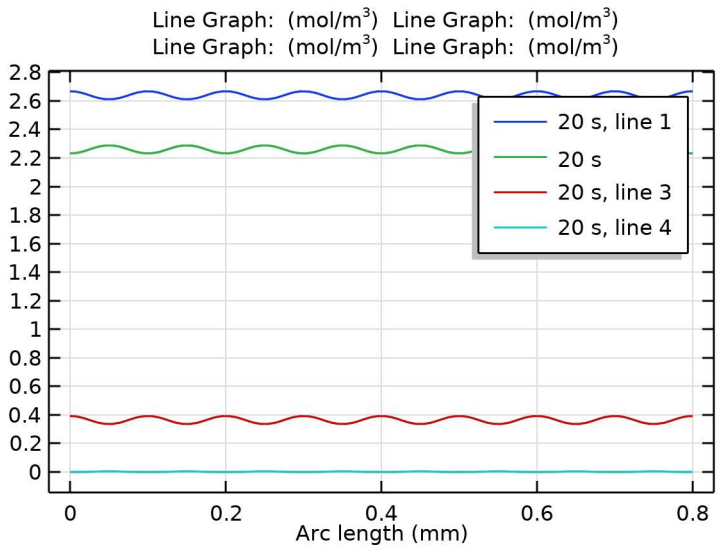
Surface: Concentration (mol/m^3) Streamline: Total flux

4.4.2 2D Plot Group 2



Streamline: Total flux

4.4.3 1D Plot Group 3



Line Graph: (mol/m^3) Line Graph: (mol/m^3) Line Graph: (mol/m^3) Line Graph: (mol/m^3)